

Potential Shaping with Changing Coalitions: A Fixed-Population Ledger Repair

Abstract

MARS-RA uses multimodal pairwise comparisons, Bradley–Terry aggregation, and potential-based reward shaping for cooperative multi-agent reinforcement learning with changing active coalitions. Its comparison scores and potentials live in the current coalition space, so consecutive vector potentials need not have the same dimension; it also combines a scalar team reward with a vector shaping term without declaring payoff semantics. We give a narrow repair. Active credits are embedded in a fixed population space, inactive agents take null actions, and every population member retains a reward ledger on every transition. Componentwise telescoping then shifts each declared per-agent payoff by a policy-independent constant, preserving best responses and Nash equilibria for arbitrary comparison outputs. If shaping rewards are retained only while an agent is active, entry and exit boundary terms remain and may depend on policy. The general Nash-invariance principle for potential shaping is prior work; the contribution here is its typed ledger formulation for changing coalitions and the associated boundary diagnosis. We also separate Bradley–Terry sampling error from semantic bias and explain why neither comparison accuracy nor a local signal bound yields a finite-training guarantee without a learner-stability premise.

1 Introduction

Learning-augmented algorithms require a declared prediction interface, an error measure, and a theorem connecting prediction quality to downstream performance. They distinguish consistency and error-sensitive guarantees from prediction-independent robustness, and warn that semantic predictions do not automatically compose into system guarantees [3]. MARS-RA instead uses multimodal pairwise comparisons to fit Bradley–Terry contribution scores and converts their softmax into a potential-based shaping signal for cooperative multi-agent reinforcement learning [2].

The tempting question is how comparison error degrades the limiting return. That question is misplaced under exact potential shaping: the discounted shaping terms telescope, so their total is a common initial offset under the usual terminal and initial-potential conditions. The comparisons may affect optimisation, exploration, or sample efficiency, but comparison accuracy alone cannot

control finite-training return without a stability, regret, or convergence contract for the learner.

Our result is narrower. MARS-RA defines a fixed population but places each score in the current active-coalition space, then subtracts consecutive vector potentials. This expression is not typed when coalition sizes change. A fixed-population embedding is readily available, but inactive coordinates, transition ownership, and per-agent payoffs must be declared. We provide that construction and show exactly where an active-only implementation fails to telescope.

The main Nash-invariance step is not new. Lu, Schwartz, and Givigi prove that potential-based shaping preserves Nash equilibria in general-sum stochastic games [1]. Our contribution is the fixed-population reward-ledger formulation for changing coalitions, together with the observation that omitting inactive-agent ledger entries leaves potentially policy-dependent boundary terms. We do not claim a new general reward-transformation theorem.

2 What the two papers establish

The survey of Zhao et al. provides a vocabulary and discipline for prediction interfaces, robustness, error propagation, and prediction cost; it does not analyse MARS-RA or prove a MARL learner-stability theorem [3]. Wang et al. define MARS-RA over a fixed population with a changing active coalition. They obtain pairwise multimodal comparisons, aggregate them through a Bradley–Terry estimator, apply softmax, and use the resulting vector in potential-based reward shaping [2].

Their estimator statement concerns concentration around a latent Bradley–Terry preference c_t^* , not a true contribution target v_t^* . The identification $c_t^* = v_t^*$ is an additional premise. Thus repeated queries can reduce sampling error while systematic semantic bias remains. Moreover, the supplied checks report that the stated estimator rate is not ready for composition: its population-size dependence is inconsistent across statements, connectedness alone does not supply uniform curvature, an unregularised maximum-likelihood estimate may diverge under separation, and the proof contains an approximate Taylor step. We therefore use no numerical Bradley–Terry rate.

3 Fixed-population formulation

Let the population be $N = \{1, \dots, n\}$ and let $I_t \subseteq N$ be the active coalition. Let $\hat{c}_t$ be any realised score vector indexed by I_t . We permit biased, stochastic, or history-dependent scores. Any information needed to make the realised potential a state function may be included in an augmented state. This qualification is necessary for a standard stochastic-game equilibrium interpretation, although the algebra below is pathwise.

For nonterminal t with $I_t \neq \emptyset$, define $\Phi_t \in \mathbb{R}^n$ by

$$\Phi_t^i = \mathbf{1}\{i \in I_t\} \text{softmax}_{I_t}(\hat{c}_t)^i, \quad i \in N, \quad (1)$$

and set $\Phi_T = 0$. A complete model must separately declare a fallback for an empty coalition; nothing below supplies one. Give inactive agents null actions and maintain a payoff ledger for every $i \in N$ on every transition. For declared per-agent environmental payoff r_t^i , define the repaired team reward and objective by

$$r_t^{\text{team}} = \sum_{i \in N} r_t^i, \quad J^{\text{team}}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t^{\text{team}} \right]. \quad (2)$$

If these payoffs allocate MARS-RA's original scalar team reward, this definition additionally requires their sum to equal that reward on each transition. Define

$$F_t^i = \gamma \Phi_{t+1}^i - \Phi_t^i, \quad \tilde{r}_t^i = r_t^i + \rho F_t^i. \quad (3)$$

Theorem 1 (fixed-population ledger invariance). *For a fixed initial augmented state and terminal potential $\Phi_T = 0$, the construction in (1)–(3) shifts player i 's discounted payoff by $-\rho \Phi_0^i$. Consequently, it preserves every unilateral payoff difference, best-response correspondence, and Nash equilibrium. It also preserves the ordering of joint policies under the repaired team objective J^{team} in (2). These conclusions hold for arbitrary realised comparison outputs on a common pathwise realisation.*

Proof. For each $i \in N$,

$$\begin{aligned} \sum_{t=0}^{T-1} \gamma^t F_t^i &= \sum_{t=0}^{T-1} (\gamma^{t+1} \Phi_{t+1}^i - \gamma^t \Phi_t^i) \\ &= -\Phi_0^i + \gamma^T \Phi_T^i = -\Phi_0^i. \end{aligned}$$

The initial augmented state is fixed, so the final quantity is independent of every player's policy. Adding a constant to player i 's payoff preserves all unilateral payoff differences and hence its best responses. Applying this to every player preserves Nash equilibria. Summing the componentwise identities gives the team-policy statement. $\square$

This theorem is a proposed repair, not a theorem stated by MARS-RA. Its general equilibrium-invariance logic is an instance of the established potential-shaping result for stochastic games [1]; what is specific here is the fixed index space and the all-population transition ledger.

4 The active-only boundary

Suppose instead that player i retains shaping payments only during an active interval $[a, b)$. Direct summation gives boundary terms of the form

$$\sum_{t=a}^{b-1} \gamma^t F_t^i = -\gamma^a \Phi_a^i + \gamma^b \Phi_b^i, \quad (4)$$

with the second term adjusted according to ownership of the exit transition. Entry, exit, and coalition changes can depend on earlier actions, so (4) need not be policy-independent. Active-only equivalence therefore requires explicit transition ownership or compensating transfers that reproduce the fixed ledger sum. The inputs do not specify those transfers, so we leave their design open.

There is a second interface boundary. MARS-RA begins from a scalar team reward but adds a vector shaping term. One coherent repair declares per-agent payoffs and applies the componentwise theorem. Another uses a scalar team potential. However, summing the embedded softmax credits gives $\sum_i \Phi_t^i = 1$ for every nonempty nonterminal coalition, so this direct scalarisation erases all comparison information. Any comparison-dependent effect must therefore occur through redistributed per-agent finite-training signals, not an added team return.

5 Estimation, semantics, and local signal sensitivity

After fixing the translation gauge of Bradley–Terry scores, define semantic bias

$$\delta_t = \|c_t^* - v_t^*\|_2. \quad (5)$$

Then

$$\|\hat{c}_t - v_t^*\|_2 \leq \|\hat{c}_t - c_t^*\|_2 + \delta_t. \quad (6)$$

Equation (6) is only the triangle inequality. It separates sampling error about the comparator’s latent preference from semantic error to the declared contribution target; it does not validate that target or repair the estimator theorem.

For this local comparison, assume that the estimated and target scores at each time are indexed by the same nonempty active coalition I_t . Let

$$d_t = \hat{c}_t - v_t^* \in \mathbb{R}^{|I_t|}, \quad (7)$$

after fixing a common translation gauge. When it appears in the shaping-signal bound below, regard d_t as zero-coordinate embedded in $\mathbb{R}^n$, and set $d_T = 0 \in \mathbb{R}^n$ by convention. Embed the target credits in the same fixed population space by

$$(\Phi_t^*)^i = \mathbf{1}\{i \in I_t\} \text{softmax}_{I_t}(v_t^*)^i, \quad \Phi_T^* = 0, \quad (8)$$

and define the vector shaping signals

$$\hat{F}_t = \gamma \Phi_{t+1} - \Phi_t, \quad F_t^* = \gamma \Phi_{t+1}^* - \Phi_t^*. \quad (9)$$

The zero-coordinate embeddings preserve Euclidean distances on their common active coalition. Since softmax is 1/2-Lipschitz in Euclidean norm, these one-step shaping signals satisfy

$$\|\hat{F}_t - F_t^*\|_2 \leq \frac{1}{2} (\gamma \|d_{t+1}\|_2 + \|d_t\|_2). \quad (10)$$

This is a local reward-signal bound only. It does not bound the return of a policy produced by MAPPO after finitely many updates. Learners can react differently to the same shaped signal, and neither source paper provides the missing stability, regret, or convergence premise.

6 Related work

Lu, Schwartz, and Givigi extend potential-based shaping to multi-player general-sum stochastic games and prove preservation of Nash equilibria [1]. Their paper also reports similar contemporaneous work by Devlin and Kudenko. Accordingly, the telescoping and general Nash-invariance argument is already published and is not a contribution of this paper. Our literature search did not locate a verified publication that states the particular fixed-population, inactive-coordinate, all-transition ledger construction for changing active coalitions. This is a search result, not a novelty proof.

The two focal papers supply the remaining context. Zhao et al. insist on typed semantic interfaces and explicit downstream composition [3]; Wang et al. supply the changing-coalition comparison and shaping mechanism to which the interface audit applies [2].

7 Interpretation

The repair makes objective-level robustness prediction-independent: once payoff semantics and ledger ownership are fixed, even inaccurate comparisons preserve payoff differences. This should not be read as evidence that comparisons help learning. Rather, it identifies the only supported location of any benefit: the finite-training optimisation path induced by redistributed per-agent signals. Better estimation can reduce one component of signal error while sharpening a semantically biased target.

8 Limitations and open questions

The result is algebraic and conditional on declared per-agent payoffs, a fixed initial augmented state, a terminal zero potential, null inactive actions, and an all-population transition ledger. It is undefined when $I_t = \emptyset$ unless the model adds an explicit empty-coalition potential convention. It does not establish that the published MARS-RA implementation uses this ledger. It does not give compensating entry or exit transfers for active-only storage. For stochastic or history-dependent comparison outputs, an equilibrium theorem may require state augmentation beyond the pathwise identity.

We do not provide a corrected Bradley–Terry concentration rate, a MAPPO stability theorem, a finite-training return bound, or a query-cost trade-off. These remain open, as do common units for pricing multimodal-model queries against return or sample efficiency. The fixed-space theorem is a correctness

layer, not evidence of a positive performance gain and not a general theory of open Markov games.

9 Conclusion

Changing coalitions expose a typing and accounting requirement hidden by the usual potential-shaping notation. A fixed-population all-transition ledger restores componentwise telescoping and preserves payoff differences for arbitrary comparison outputs. Active-only accounting leaves boundary terms, while direct scalarisation removes the comparison signal. These conclusions narrow the contribution of comparison-based shaping: its supported role is in per-agent finite-training dynamics, for which a learner contract is still missing.

References

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